Core Java Practical Slips – 1 to 30

Only Core Java Q1 programs are included. Web Technology and Viva questions are omitted.

Slip No. 1

Question: Write a Java program to print the sum of elements of the array. Also display array elements in ascending order.

Code:

```
import java.util.*;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter size: ");
        int n = sc.nextInt();
        int[] a = new int[n];

        int sum = 0;
        for (int i = 0; i < n; i++) {
            a[i] = sc.nextInt();
            sum += a[i];
        }

        Arrays.sort(a);

        System.out.println("Sum = " + sum);
        System.out.println("Ascending order:");
        for (int x : a)
            System.out.print(x + " ");
    }
}
```

Slip No. 2

Question: Write a Java Program to Display Armstrong Numbers Between range. Accept range from user.

Code:

```
import java.util.*;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter range: ");
        int start = sc.nextInt();
        int end = sc.nextInt();

        for (int n = start; n <= end; n++) {
            int temp = n, sum = 0;

            while (temp > 0) {
                int d = temp % 10;
                sum += d * d * d;
                temp /= 10;
            }

            if (sum == n)
                System.out.print(n + " ");
        }
    }
}
```

Slip No. 3

Question: Write a package for String operation which has two classes Con and Comp. Con class has to concatenate two strings and Comp class compares two strings. Also display proper messages on execution.

Code:

```
// Con.java
package stringop;

public class Con {
    public void concatenate(String a, String b) {
        System.out.println("Concatenated String: " + a + b);
    }
}

// Comp.java
package stringop;

public class Comp {
    public void compare(String a, String b) {
        if (a.equals(b))
            System.out.println("Strings are equal");
        else
            System.out.println("Strings are not equal");
    }
}

// Main.java
import stringop.*;

class Main {
    public static void main(String[] args) {
        Con c = new Con();
        Comp p = new Comp();

        c.concatenate("Hello ", "Java");
        p.compare("Java", "Java");
    }
}
```

Slip No. 4

Question: Write a menu driven program to perform the following operations on multidimensional array ie matrix: Addition, Multiplication, Transpose of any matrix, Exit.

Code:

```
import java.util.*;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        while (true) {
            System.out.println("\n1.Addition  2.Multiplication  3.Transpose  4.Exit");
            int ch = sc.nextInt();

            if (ch == 4)
                break;

            System.out.print("Enter rows and columns: ");
            int r = sc.nextInt();
            int c = sc.nextInt();

            int[][] a = new int[r][c];

            for (int i = 0; i < r; i++)
                for (int j = 0; j < c; j++)
                    a[i][j] = sc.nextInt();

            if (ch == 1) {
                int[][] b = new int[r][c];
                int[][] sum = new int[r][c];

                for (int i = 0; i < r; i++)
                    for (int j = 0; j < c; j++)
                        b[i][j] = sc.nextInt();

                for (int i = 0; i < r; i++)
                    for (int j = 0; j < c; j++)
                        sum[i][j] = a[i][j] + b[i][j];

                for (int[] row : sum)
                    System.out.println(Arrays.toString(row));

            } else if (ch == 2) {
                System.out.print("Enter columns of second matrix: ");
                int c2 = sc.nextInt();
                int[][] b = new int[c][c2];

                for (int i = 0; i < c; i++)
                    for (int j = 0; j < c2; j++)
                        b[i][j] = sc.nextInt();

                int[][] p = new int[r][c2];

                for (int i = 0; i < r; i++)
                    for (int j = 0; j < c2; j++)
                        for (int k = 0; k < c; k++)
                            p[i][j] += a[i][k] * b[k][j];

                for (int[] row : p)
                    System.out.println(Arrays.toString(row));

            } else if (ch == 3) {
                for (int j = 0; j < c; j++) {
                    for (int i = 0; i < r; i++)
                        System.out.print(a[i][j] + " ");
                    System.out.println();
                }
            }
        }
    }
}
```

Slip No. 5

Question: Write a program to accept a text file from user and display the contents of a file in reverse order and change its case.

Code:

```
import java.io.*;
import java.util.*;

class Main {
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter file name: ");
        String file = sc.nextLine();

        BufferedReader br = new BufferedReader(new FileReader(file));
        StringBuilder s = new StringBuilder();
        String line;

        while ((line = br.readLine()) != null)
            s.append(line);

        br.close();

        String text = s.reverse().toString();

        for (char ch : text.toCharArray()) {
            if (Character.isUpperCase(ch))
                System.out.print(Character.toLowerCase(ch));
            else if (Character.isLowerCase(ch))
                System.out.print(Character.toUpperCase(ch));
            else
                System.out.print(ch);
        }
    }
}
```

Slip No. 6

Question: Write a program to define a class Account having members custname, accno. Define default and parameterized constructor. Create a subclass called SavingAccount with members savingbal, minbal. Create a derived class AccountDetail that extends the class SavingAccount with members depositamt and withdrawalamt. Write an appropriate method to display customer details.

Code:

```
class Account {
    String custname;
    int accno;

    Account() {
        custname = "Unknown";
        accno = 0;
    }

    Account(String name, int no) {
        custname = name;
        accno = no;
    }
}

class SavingAccount extends Account {
    double savingbal, minbal;

    SavingAccount(String name, int no, double bal, double min) {
        super(name, no);
        savingbal = bal;
        minbal = min;
    }
}

class AccountDetail extends SavingAccount {
    double depositamt, withdrawalamt;

    AccountDetail(String name, int no, double bal, double min,
                  double deposit, double withdrawal) {
        super(name, no, bal, min);
        depositamt = deposit;
        withdrawalamt = withdrawal;
    }

    void display() {
        System.out.println("Customer: " + custname);
        System.out.println("Account No: " + accno);
        System.out.println("Saving Balance: " + savingbal);
        System.out.println("Minimum Balance: " + minbal);
        System.out.println("Deposit: " + depositamt);
        System.out.println("Withdrawal: " + withdrawalamt);
    }
}

class Main {
    public static void main(String[] args) {
        AccountDetail a = new AccountDetail("Rahul", 101, 10000, 1000, 2000, 500);
        a.display();
    }
}
```

Slip No. 7

Question: Write a class Driver with attributes license_no, name, address and age. Initialize values through the parameterized constructor. If age of Driver is less than 18 then user-defined exception should be generated —Age is below 18 years—

Code:

```
class AgeException extends Exception {
    AgeException(String msg) {
        super(msg);
    }
}

class Driver {
    String license_no, name, address;
    int age;

    Driver(String l, String n, String a, int age) throws AgeException {
        if (age < 18)
            throw new AgeException("Age is below 18 years");

        license_no = l;
        name = n;
        address = a;
        this.age = age;
    }
}

class Main {
    public static void main(String[] args) {
        try {
            Driver d = new Driver("MH12A1234", "Rahul", "Pune", 20);
            System.out.println("Driver created successfully");
        } catch (AgeException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

Slip No. 8

Question: Write a program to create an abstract class named Shape that contains two integers and an empty method named printArea(). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contain only the method printArea() that prints the area of the given shape. (use method overriding).

Code:

```
abstract class Shape {
    int a, b;

    abstract void printArea();
}

class Rectangle extends Shape {
    Rectangle(int l, int w) {
        a = l;
        b = w;
    }

    void printArea() {
        System.out.println("Rectangle Area = " + (a * b));
    }
}

class Triangle extends Shape {
    Triangle(int base, int height) {
        a = base;
        b = height;
    }

    void printArea() {
        System.out.println("Triangle Area = " + (0.5 * a * b));
    }
}

class Circle extends Shape {
    Circle(int r) {
        a = r;
    }

    void printArea() {
        System.out.println("Circle Area = " + (Math.PI * a * a));
    }
}

class Main {
    public static void main(String[] args) {
        new Rectangle(10, 5).printArea();
        new Triangle(10, 5).printArea();
        new Circle(5).printArea();
    }
}
```

Slip No. 9

Question: Write a program which define class Employee with data member as id, name and salary. Store the information of 'n' employees and display the name of employee having maximum salary (Use array of object).

Code:

```
import java.util.*;

class Employee {
    int id;
    String name;
    double salary;

    Employee(int id, String name, double salary) {
        this.id = id;
        this.name = name;
        this.salary = salary;
    }
}

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of employees: ");
        int n = sc.nextInt();

        Employee[] e = new Employee[n];

        for (int i = 0; i < n; i++) {
            System.out.print("Enter id, name and salary: ");
            int id = sc.nextInt();
            String name = sc.next();
            double salary = sc.nextDouble();
            e[i] = new Employee(id, name, salary);
        }

        Employee max = e[0];

        for (int i = 1; i < n; i++)
            if (e[i].salary > max.salary)
                max = e[i];

        System.out.println("Employee with maximum salary: " + max.name);
    }
}
```

Slip No. 10

Question: Write a program to read the contents of “abc.txt” file. Display the contents of file in uppercase as output.

Code:

```
import java.io.*;

class Main {
    public static void main(String[] args) throws Exception {
        BufferedReader br = new BufferedReader(new FileReader("abc.txt"));
        String line;

        while ((line = br.readLine()) != null)
            System.out.println(line.toUpperCase());

        br.close();
    }
}
```

Slip No. 11

Question: Write a program to create a super class Vehicle having members Company and price. Derive two different classes LightMotorVehicle (mileage) and HeavyMotorVehicle (capacity_in_tons). Accept the information for "n" vehicles and display the information in appropriate form. While taking data, ask user about the type of vehicle first.

Code:

```
import java.util.*;

class Vehicle {
    String company;
    double price;

    Vehicle(String company, double price) {
        this.company = company;
        this.price = price;
    }
}

class LightMotorVehicle extends Vehicle {
    double mileage;

    LightMotorVehicle(String company, double price, double mileage) {
        super(company, price);
        this.mileage = mileage;
    }

    void display() {
        System.out.println(company + " " + price + " " + mileage);
    }
}

class HeavyMotorVehicle extends Vehicle {
    double capacity_in_tons;

    HeavyMotorVehicle(String company, double price, double capacity) {
        super(company, price);
        capacity_in_tons = capacity;
    }

    void display() {
        System.out.println(company + " " + price + " " + capacity_in_tons);
    }
}

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of vehicles: ");
        int n = sc.nextInt();

        for (int i = 0; i < n; i++) {
            System.out.print("Enter type (1-Light, 2-Heavy): ");
            int type = sc.nextInt();

            System.out.print("Company and price: ");
            String company = sc.next();
            double price = sc.nextDouble();

            if (type == 1) {
                System.out.print("Mileage: ");
                double mileage = sc.nextDouble();
                new LightMotorVehicle(company, price, mileage).display();
            } else {
                System.out.print("Capacity in tons: ");
                double capacity = sc.nextDouble();
                new HeavyMotorVehicle(company, price, capacity).display();
            }
        }
    }
}
```

Slip No. 12

Question: Write a package game which will have 2 classes Indoor & Outdoor. Use a function display() to generate the list of players for the specific game. Use default & parameterized constructor.

Code:

```
// Indoor.java
package game;

public class Indoor {
    String game;

    public Indoor() {
        game = "Chess";
    }

    public Indoor(String game) {
        this.game = game;
    }

    public void display() {
        System.out.println("Indoor Game: " + game);
    }
}

// Outdoor.java
package game;

public class Outdoor {
    String game;

    public Outdoor() {
        game = "Cricket";
    }

    public Outdoor(String game) {
        this.game = game;
    }

    public void display() {
        System.out.println("Outdoor Game: " + game);
    }
}

// Main.java
import game.*;

class Main {
    public static void main(String[] args) {
        Indoor i = new Indoor("Chess");
        Outdoor o = new Outdoor("Cricket");

        i.display();
        o.display();
    }
}
```

Slip No. 13

Question: Define a “Clock” class that does the following: Accept Hours, Minutes and Seconds; Check the validity of numbers; Set the time to AM/PM mode. Use the necessary constructors and methods to do the above task.

Code:

```
import java.util.*;

class Clock {
    int h, m, s;

    Clock(int h, int m, int s) {
        this.h = h;
        this.m = m;
        this.s = s;
    }

    boolean valid() {
        return h >= 0 && h <= 23 && m >= 0 && m <= 59 && s >= 0 && s <= 59;
    }

    void display() {
        int hour = h;
        String ampm = hour >= 12 ? "PM" : "AM";

        hour = hour % 12;
        if (hour == 0)
            hour = 12;

        System.out.printf("%02d:%02d:%02d %s", hour, m, s, ampm);
    }
}

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int h = sc.nextInt();
        int m = sc.nextInt();
        int s = sc.nextInt();

        Clock c = new Clock(h, m, s);

        if (c.valid())
            c.display();
        else
            System.out.println("Invalid Time");
    }
}
```

Slip No. 14

Question: Write a Java program to print the sum of elements of the array. Also display array elements in ascending order.

Code:

```
import java.util.*;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] a = new int[n];
        int sum = 0;

        for (int i = 0; i < n; i++) {
            a[i] = sc.nextInt();
            sum += a[i];
        }

        Arrays.sort(a);

        System.out.println("Sum = " + sum);
        for (int x : a)
            System.out.print(x + " ");
    }
}
```

Slip No. 15

Question: Write a program for multilevel inheritance such that Country is inherited from Continent. State is inherited from Country. Display the place, State, Country and Continent.

Code:

```
class Continent {
    String continent = "Asia";
}

class Country extends Continent {
    String country = "India";
}

class State extends Country {
    String state = "Maharashtra";
    String place = "Pune";

    void display() {
        System.out.println("Place: " + place);
        System.out.println("State: " + state);
        System.out.println("Country: " + country);
        System.out.println("Continent: " + continent);
    }
}

class Main {
    public static void main(String[] args) {
        State s = new State();
        s.display();
    }
}
```

Slip No. 16

Question: Create an abstract class Shape with methods area & volume. Derive a class Cylinder (radius, height). Calculate area and volume.

Code:

```
abstract class Shape {
    abstract void area();
    abstract void volume();
}

class Cylinder extends Shape {
    double r, h;

    Cylinder(double r, double h) {
        this.r = r;
        this.h = h;
    }

    void area() {
        System.out.println("Surface Area = " + (2 * Math.PI * r * (r + h)));
    }

    void volume() {
        System.out.println("Volume = " + (Math.PI * r * r * h));
    }
}

class Main {
    public static void main(String[] args) {
        Cylinder c = new Cylinder(5, 10);
        c.area();
        c.volume();
    }
}
```

Slip No. 17

Question: Write a Java program that defines an interface Calculator containing methods add() and subtract() and implements it in class SimpleCalc.

Code:

```
interface Calculator {
    void add(int a, int b);
    void subtract(int a, int b);
}

class SimpleCalc implements Calculator {
    public void add(int a, int b) {
        System.out.println("Addition = " + (a + b));
    }

    public void subtract(int a, int b) {
        System.out.println("Subtraction = " + (a - b));
    }
}

class Main {
    public static void main(String[] args) {
        SimpleCalc c = new SimpleCalc();
        c.add(10, 5);
        c.subtract(10, 5);
    }
}
```

Slip No. 18

Question: Define a class MyNumber having one private integer data member. Write a default constructor initialize it to 0 and another constructor to initialize it to a value. Write methods isNegative, isPositive, isOdd, isEven. Use command line argument to pass a value to the object and perform the above operations.

Code:

```
class MyNumber {
    private int n;

    MyNumber() {
        n = 0;
    }

    MyNumber(int n) {
        this.n = n;
    }

    void isNegative() {
        System.out.println("Negative: " + (n < 0));
    }

    void isPositive() {
        System.out.println("Positive: " + (n > 0));
    }

    void isOdd() {
        System.out.println("Odd: " + (n % 2 != 0));
    }

    void isEven() {
        System.out.println("Even: " + (n % 2 == 0));
    }
}

class Main {
    public static void main(String[] args) {
        int n = Integer.parseInt(args[0]);

        MyNumber x = new MyNumber(n);
        x.isNegative();
        x.isPositive();
        x.isOdd();
        x.isEven();
    }
}
```

Slip No. 19

Question: Write a Java program to accept a file name from the user and count the number of characters, words, and lines present in the file.

Code:

```
import java.io.*;
import java.util.*;

class Main {
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter file name: ");
        String name = sc.nextLine();

        BufferedReader br = new BufferedReader(new FileReader(name));

        int chars = 0, words = 0, lines = 0;
        String line;

        while ((line = br.readLine()) != null) {
            lines++;
            chars += line.length();

            String[] w = line.trim().split("\\s+");
            if (!line.trim().isEmpty())
                words += w.length;
        }

        br.close();

        System.out.println("Characters: " + chars);
        System.out.println("Words: " + words);
        System.out.println("Lines: " + lines);
    }
}
```

Slip No. 20

Question: Write a program to create parent class College (cno, cname, caddr) and derived class Department (dno, dname) from College. Write necessary methods to display College details.

Code:

```
class College {
    int cno;
    String cname, caddr;

    College(int cno, String cname, String caddr) {
        this.cno = cno;
        this.cname = cname;
        this.caddr = caddr;
    }

    void displayCollege() {
        System.out.println("College No: " + cno);
        System.out.println("College Name: " + cname);
        System.out.println("College Address: " + caddr);
    }
}

class Department extends College {
    int dno;
    String dname;

    Department(int cno, String cname, String caddr, int dno, String dname) {
        super(cno, cname, caddr);
        this.dno = dno;
        this.dname = dname;
    }
}

class Main {
    public static void main(String[] args) {
        Department d = new Department(1, "ABC College", "Pune", 10, "Computer Science");
        d.displayCollege();
    }
}
```

Slip No. 21

Question: Write a program to create parent class College (cno, cname, caddr) and derived class Department (dno, dname) from College. Write necessary methods to display College details.

Code:

```
class College {
    int cno;
    String cname, caddr;

    College(int cno, String cname, String caddr) {
        this.cno = cno;
        this.cname = cname;
        this.caddr = caddr;
    }

    void display() {
        System.out.println(cno);
        System.out.println(cname);
        System.out.println(caddr);
    }
}

class Department extends College {
    int dno;
    String dname;

    Department(int cno, String cname, String caddr, int dno, String dname) {
        super(cno, cname, caddr);
        this.dno = dno;
        this.dname = dname;
    }
}

class Main {
    public static void main(String[] args) {
        Department d = new Department(1, "ABC College", "Pune", 10, "Computer Science");
        d.display();
    }
}
```

Slip No. 22

Question: Write a program using interface, create a class Product (product_id, product_name, product_cost, product_quantity) default and parameterized constructor. Create objects of class product and display the contents of each object and also display the object count.

Code:

```
interface ProductInfo {
    void display();
}

class Product implements ProductInfo {
    int product_id, product_quantity;
    String product_name;
    double product_cost;
    static int count = 0;

    Product() {
        product_id = 0;
        product_name = "Unknown";
        product_cost = 0;
        product_quantity = 0;
        count++;
    }

    Product(int id, String name, double cost, int quantity) {
        product_id = id;
        product_name = name;
        product_cost = cost;
        product_quantity = quantity;
        count++;
    }

    public void display() {
        System.out.println(product_id + " " + product_name + " " +
                           product_cost + " " + product_quantity);
    }
}

class Main {
    public static void main(String[] args) {
        Product p1 = new Product();
        Product p2 = new Product(1, "Pen", 10, 5);

        p1.display();
        p2.display();

        System.out.println("Object Count: " + Product.count);
    }
}
```

Slip No. 23

Question: Write a program for multilevel inheritance such that Country is inherited from Continent. State is inherited from Country. Display the place, State, Country and Continent.

Code:

```
class Continent {
    String continent = "Asia";
}

class Country extends Continent {
    String country = "India";
}

class State extends Country {
    String state = "Maharashtra";
    String place = "Pune";

    void display() {
        System.out.println("Place: " + place);
        System.out.println("State: " + state);
        System.out.println("Country: " + country);
        System.out.println("Continent: " + continent);
    }
}

class Main {
    public static void main(String[] args) {
        new State().display();
    }
}
```

Slip No. 24

Question: Create an abstract class Shape with methods area & volume. Derive a class Cylinder (radius, height). Calculate area and volume.

Code:

```
abstract class Shape {
    abstract void area();
    abstract void volume();
}

class Cylinder extends Shape {
    double radius, height;

    Cylinder(double radius, double height) {
        this.radius = radius;
        this.height = height;
    }

    void area() {
        System.out.println("Area = " +
            (2 * Math.PI * radius * (radius + height)));
    }

    void volume() {
        System.out.println("Volume = " +
            (Math.PI * radius * radius * height));
    }
}

class Main {
    public static void main(String[] args) {
        Cylinder c = new Cylinder(5, 10);
        c.area();
        c.volume();
    }
}
```

Slip No. 25

Question: Write a Java program that defines an interface Calculator containing methods add() and subtract() and implements it in class SimpleCalc.

Code:

```
interface Calculator {
    void add(int a, int b);
    void subtract(int a, int b);
}

class SimpleCalc implements Calculator {
    public void add(int a, int b) {
        System.out.println("Addition = " + (a + b));
    }

    public void subtract(int a, int b) {
        System.out.println("Subtraction = " + (a - b));
    }
}

class Main {
    public static void main(String[] args) {
        SimpleCalc c = new SimpleCalc();
        c.add(10, 5);
        c.subtract(10, 5);
    }
}
```

Slip No. 26

Question: Define a class Student with data members student_name, student_rollno, total_lectures, attended_lectures. Create an object of Student. If attendance percentage is less than 75%, throw a user-defined exception “Student is Not Eligible for Exam”, otherwise display student details.

Code:

```
class AttendanceException extends Exception {
    AttendanceException(String msg) {
        super(msg);
    }
}

class Student {
    String student_name;
    int student_rollno, total_lectures, attended_lectures;

    Student(String name, int roll, int total, int attended) {
        student_name = name;
        student_rollno = roll;
        total_lectures = total;
        attended_lectures = attended;
    }

    void display() throws AttendanceException {
        double percentage =
            (attended_lectures * 100.0) / total_lectures;

        if (percentage < 75)
            throw new AttendanceException(
                "Student is Not Eligible for Exam");

        System.out.println("Name: " + student_name);
        System.out.println("Roll No: " + student_rollno);
        System.out.println("Attendance: " + percentage + "%");
    }
}

class Main {
    public static void main(String[] args) {
        try {
            Student s = new Student("Rahul", 1, 100, 80);
            s.display();
        } catch (AttendanceException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

Slip No. 27

Question: Write a program to define class Person with data members Personname, Aadharno, Panno. Accept information for 5 objects and display appropriate information (use this keyword).

Code:

```
import java.util.*;

class Person {
    String Personname, Aadharno, Panno;

    Person(String Personname, String Aadharno, String Panno) {
        this.Personname = Personname;
        this.Aadharno = Aadharno;
        this.Panno = Panno;
    }

    void display() {
        System.out.println(Personname + " " + Aadharno + " " + Panno);
    }
}

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Person[] p = new Person[5];

        for (int i = 0; i < 5; i++) {
            String name = sc.next();
            String aadhar = sc.next();
            String pan = sc.next();

            p[i] = new Person(name, aadhar, pan);
        }

        for (Person x : p)
            x.display();
    }
}
```

Slip No. 28

Question: Define a class MyDate(Day, Month, year) with methods to accept and display a MyDate object. Accept date as dd,mm,yyyy. Throw user defined exception "InvalidDateException" if the date is invalid.

Code:

```
import java.util.*;

class InvalidDateException extends Exception {
    InvalidDateException(String msg) {
        super(msg);
    }
}

class MyDate {
    int day, month, year;

    void accept(int d, int m, int y) throws InvalidDateException {
        if (m < 1 || m > 12 || d < 1 || d > 31)
            throw new InvalidDateException("InvalidDateException");

        day = d;
        month = m;
        year = y;
    }

    void display() {
        System.out.println(day + "/" + month + "/" + year);
    }
}

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int d = sc.nextInt();
        int m = sc.nextInt();
        int y = sc.nextInt();

        try {
            MyDate date = new MyDate();
            date.accept(d, m, y);
            date.display();
        } catch (InvalidDateException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

Slip No. 29

Question: Write a program to accept a number from the user, if number is zero then throw user defined exception “Number is 0” otherwise check whether no is prime or not (Use static keyword).

Code:

```
import java.util.*;

class NumberException extends Exception {
    NumberException(String msg) {
        super(msg);
    }
}

class Main {
    static void checkPrime(int n) throws NumberException {
        if (n == 0)
            throw new NumberException("Number is 0");

        boolean prime = true;

        if (n < 2)
            prime = false;

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                prime = false;
                break;
            }
        }

        if (prime)
            System.out.println("Prime");
        else
            System.out.println("Not Prime");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            checkPrime(sc.nextInt());
        } catch (NumberException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

Slip No. 30

Question: Define a class MyNumber having one private integer data member. Write a default constructor initialize it to 0 and another constructor to initialize it to a value. Write methods isNegative, isPositive, isOdd, isEven. Use command line argument to pass a value to the object and perform the above operations.

Code:

```
class MyNumber {
    private int n;

    MyNumber() {
        n = 0;
    }

    MyNumber(int n) {
        this.n = n;
    }

    void isNegative() {
        System.out.println("Negative: " + (n < 0));
    }

    void isPositive() {
        System.out.println("Positive: " + (n > 0));
    }

    void isOdd() {
        System.out.println("Odd: " + (n % 2 != 0));
    }

    void isEven() {
        System.out.println("Even: " + (n % 2 == 0));
    }
}

class Main {
    public static void main(String[] args) {
        MyNumber n = new MyNumber(Integer.parseInt(args[0]));

        n.isNegative();
        n.isPositive();
        n.isOdd();
        n.isEven();
    }
}
```